Judson, *Abstract Algebra: Theory and Applications* Chapter 11 (Homomorphisms): Solutions to Exercises 1–20 and Additional Exercises (Automorphisms)

Exercise 1

Prove that $\det(AB) = \det(A)\det(B)$ for $A, B \in \mathrm{GL}_2(\mathbb{R})$. This shows that the determinant is a homomorphism from $\mathrm{GL}_2(\mathbb{R})$ to $\mathbb{R}^*$.

Solution

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$. Then

$$AB = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix},$$

so

$$\begin{aligned} \det(AB) &= (ae+bg)(cf+dh) - (af+bh)(ce+dg) \\ &= acef + adeh + bcfg + bdgh - acef - adfg - bceh - bdgh \\ &= adeh + bcfg - adfg - bceh \\ &= ad(eh-fg) - bc(eh-fg) \\ &= (ad-bc)(eh-fg) = \det(A)\det(B). \end{aligned}$$

Since A and B are invertible, $\det A$ and $\det B$ are nonzero, so det maps $\mathrm{GL}_2(\mathbb{R})$ into $\mathbb{R}^*$, and the identity above says it preserves the group operation. □

Exercise 2

Which of the following maps are homomorphisms? If the map is a homomorphism, what is the kernel?

(a) $\phi : \mathbb{R}^* \to \mathrm{GL}_2(\mathbb{R})$ defined by $\phi(a) = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}$

(b) $\phi : \mathbb{R} \to \mathrm{GL}_2(\mathbb{R})$ defined by $\phi(a) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$

(c) $\phi : \mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}$ defined by $\phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d$

(d) $\phi : \mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^*$ defined by $\phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$

(e) $\phi : M_2(\mathbb{R}) \to \mathbb{R}$ defined by $\phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = b$, where $M_2(\mathbb{R})$ is the additive group of 2×2 matrices with entries in $\mathbb{R}$.

Solution

(a) Homomorphism. For $a, b \in \mathbb{R}^*$,

$$\phi(a)\phi(b) = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & ab \end{pmatrix} = \phi(ab).$$

$\phi(a) = I$ if and only if $a = 1$, so $\ker \phi = \{1\}$.

(b) Homomorphism (the group operation on $\mathbb{R}$ is addition). For $a, b \in \mathbb{R}$,

$$\phi(a)\phi(b) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a+b & 1 \end{pmatrix} = \phi(a+b).$$

$\phi(a) = I$ if and only if $a = 0$, so $\ker \phi = \{0\}$.

(c) Not a homomorphism (here $\mathbb{R}$ is a group under addition, so we would need $\phi(AB) = \phi(A) + \phi(B)$). Take $A = B = I$. Then $\phi(AB) = \phi(I) = 2$, but $\phi(A) + \phi(B) = 2 + 2 = 4$.

(d) Homomorphism, by Exercise 1. The kernel is

$$\ker \phi = \{A \in \mathrm{GL}_2(\mathbb{R}) : \det A = 1\} = \mathrm{SL}_2(\mathbb{R}),$$

the special linear group.

(e) Homomorphism. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $A' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$,

$$\phi(A + A') = b + b' = \phi(A) + \phi(A').$$

The kernel is

$$\ker \phi = \left\{ \begin{pmatrix} a & 0 \\ c & d \end{pmatrix} : a, c, d \in \mathbb{R} \right\}.$$

Exercise 3

Let A be an $m \times n$ matrix. Show that matrix multiplication, $x \mapsto Ax$, defines a homomorphism $\phi : \mathbb{R}^n \to \mathbb{R}^m$.

Solution

Regard $\mathbb{R}^n$ and $\mathbb{R}^m$ as additive groups of column vectors. For $x, y \in \mathbb{R}^n$, the distributive law for matrix multiplication gives

$$\phi(x + y) = A(x + y) = Ax + Ay = \phi(x) + \phi(y).$$

So ϕ is a homomorphism. Its kernel is the null space $\{x \in \mathbb{R}^n : Ax = 0\}$ and its image is the column space of A. □

Exercise 4

Let $\phi : \mathbb{Z} \to \mathbb{Z}$ be given by $\phi(n) = 7n$. Prove that ϕ is a group homomorphism. Find the kernel and the image of ϕ.

Solution

For $m, n \in \mathbb{Z}$, $\phi(m+n) = 7(m+n) = 7m + 7n = \phi(m) + \phi(n)$, so ϕ is a homomorphism.

If $\phi(n) = 7n = 0$, then $n = 0$, so $\ker \phi = \{0\}$. The image is $\phi(\mathbb{Z}) = \{7n : n \in \mathbb{Z}\} = 7\mathbb{Z}$.

Exercise 5

Describe all of the homomorphisms from $\mathbb{Z}_{24}$ to $\mathbb{Z}_{18}$.

Solution

A homomorphism $\phi : \mathbb{Z}_{24} \to \mathbb{Z}_{18}$ is determined by $a = \phi(1)$, since $\phi(n) = n\,a$. For this to be well defined we need $\phi(24) = \phi(0) = 0$, that is,

$$24a \equiv 0 \pmod{18} \iff 6a \equiv 0 \pmod{18} \iff a \equiv 0 \pmod{3}.$$

Conversely, if $3 \mid a$, then $\phi(n) = na \bmod 18$ is well defined (if $n \equiv n' \pmod{24}$, then $(n - n')a$ is a multiple of $24a$, which is a multiple of 18) and is a homomorphism. Hence the homomorphisms are exactly

$$\phi_a(n) = an \bmod 18, \qquad a \in \{0, 3, 6, 9, 12, 15\},$$

so there are $\gcd(24, 18) = 6$ of them. Their kernels and images:

a	$\ker \phi_a$	$\operatorname{im} \phi_a$
0	$\mathbb{Z}_{24}$	$\{0\}$
3, 15	$\langle 6 \rangle$ (order 4)	$\langle 3 \rangle$ (order 6)
6, 12	$\langle 3 \rangle$ (order 8)	$\langle 6 \rangle$ (order 3)
9	$\langle 2 \rangle$ (order 12)	$\langle 9 \rangle = \{0, 9\}$ (order 2)

Exercise 6

Describe all of the homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}_{12}$.

Solution

A homomorphism $\phi : \mathbb{Z} \to \mathbb{Z}_{12}$ is determined by $a = \phi(1) \in \mathbb{Z}_{12}$, since $\phi(n) = na \bmod 12$. There is no well-definedness condition because $\mathbb{Z}$ is free on the generator 1, so every $a \in \mathbb{Z}_{12}$ works: for any a, the map $n \mapsto na$ is a homomorphism since $(m+n)a = ma + na$. Thus there are exactly 12 homomorphisms,

$$\phi_a(n) = an \bmod 12, \qquad a = 0, 1, \ldots, 11.$$

The kernel of ϕ_a is $\{n : 12 \mid an\} = \dfrac{12}{\gcd(a, 12)}\,\mathbb{Z}$.

Exercise 7

In the group $\mathbb{Z}_{24}$, let $H = \langle 4 \rangle$ and $N = \langle 6 \rangle$.

(a) List the elements in HN (we usually write $H + N$ for these additive groups) and $H \cap N$.

(b) List the cosets in HN/N, showing the elements in each coset.

(c) List the cosets in $H/(H \cap N)$, showing the elements in each coset.

(d) Give the correspondence between HN/N and $H/(H \cap N)$ described in the proof of the Second Isomorphism Theorem.

Solution

We have

$$H = \langle 4 \rangle = \{0, 4, 8, 12, 16, 20\}, \qquad N = \langle 6 \rangle = \{0, 6, 12, 18\}.$$

(a) $H + N = \langle \gcd(4, 6) \rangle = \langle 2 \rangle$, so

$$H + N = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22\}.$$

$H \cap N = \langle \operatorname{lcm}(4, 6) \rangle = \langle 12 \rangle$, so

$$H \cap N = \{0, 12\}.$$

(b) $|HN/N| = 12/4 = 3$. The cosets are

$$\begin{aligned} 0 + N &= \{0, 6, 12, 18\}, \\ 2 + N &= \{2, 8, 14, 20\}, \\ 4 + N &= \{4, 10, 16, 22\}. \end{aligned}$$

(c) $|H/(H \cap N)| = 6/2 = 3$. The cosets are

$$\begin{aligned} 0 + (H \cap N) &= \{0, 12\}, \\ 4 + (H \cap N) &= \{4, 16\}, \\ 8 + (H \cap N) &= \{8, 20\}. \end{aligned}$$

(d) The proof of the Second Isomorphism Theorem uses the map $H \to HN/N$, $h \mapsto h + N$. It is a surjective homomorphism with kernel $H \cap N$, so it induces an isomorphism $H/(H \cap N) \to HN/N$ given by $h + (H \cap N) \mapsto h + N$. Explicitly:

$$\begin{aligned} \{0, 12\} &\longmapsto \{0, 6, 12, 18\} = 0 + N, \\ \{4, 16\} &\longmapsto \{4, 10, 16, 22\} = 4 + N, \\ \{8, 20\} &\longmapsto \{2, 8, 14, 20\} = 8 + N. \end{aligned}$$

Exercise 8

If G is an abelian group and $n \in \mathbb{N}$, show that $\phi : G \to G$ defined by $g \mapsto g^n$ is a group homomorphism.

Solution

First, in an abelian group $(gh)^n = g^n h^n$ for all $n \geq 1$. This is clear for $n = 1$, and if it holds for k, then

$$(gh)^{k+1} = (gh)^k (gh) = g^k h^k \, gh = g^k g \, h^k h = g^{k+1} h^{k+1},$$

where we moved g past h^k because G is abelian. Hence for $g, h \in G$,

$$\phi(gh) = (gh)^n = g^n h^n = \phi(g)\phi(h). \quad \square$$

Exercise 9

If $\phi : G \to H$ is a group homomorphism and G is abelian, prove that $\phi(G)$ is also abelian.

Solution

Let $x, y \in \phi(G)$, say $x = \phi(a)$ and $y = \phi(b)$. Then

$$xy = \phi(a)\phi(b) = \phi(ab) = \phi(ba) = \phi(b)\phi(a) = yx. \quad \square$$

Exercise 10

If $\phi : G \to H$ is a group homomorphism and G is cyclic, prove that $\phi(G)$ is also cyclic.

Solution

Let $G = \langle g \rangle$. Every element of $\phi(G)$ has the form $\phi(g^n) = \phi(g)^n$ for some $n \in \mathbb{Z}$, since a homomorphism satisfies $\phi(g^n) = \phi(g)^n$ for all integers n. Hence $\phi(G) = \langle \phi(g) \rangle$ is cyclic. $\square$

Exercise 11

Show that a homomorphism defined on a cyclic group is completely determined by its action on the generator of the group.

Solution

Let $G = \langle g \rangle$ and let $\phi, \psi : G \to H$ be homomorphisms with $\phi(g) = \psi(g)$. Every element of G has the form g^n, and

$$\phi(g^n) = \phi(g)^n = \psi(g)^n = \psi(g^n).$$

So $\phi = \psi$. $\square$

Exercise 12

Let G be a group of order p^2, where p is a prime number. If H is a subgroup of G of order p, show that H is normal in G. Prove that G must be abelian.

Solution

H is normal. Let G act on the set G/H of left cosets (there are p of them) by left multiplication. This gives a homomorphism $\varphi : G \to S_p$. Let $K = \ker \varphi$. If $g \in K$, then $gH = H$, so $g \in H$; thus $K \subseteq H$ and $|K| \in \{1, p\}$.

By the First Isomorphism Theorem, $G/K \cong \varphi(G) \leq S_p$, so $|G/K| = p^2/|K|$ divides $p!$. Since p divides $p!$ exactly once, $p^2 \nmid p!$, so $|G/K| \leq p$ (it is a power of p), hence $|K| \geq p$. Therefore $K = H$, and H is normal since it is a kernel.

G is abelian. If some element has order p^2, then G is cyclic, hence abelian. Otherwise every non-identity element has order p. Let $a, b \in G$ with $a \neq e$ and $b \notin \langle a \rangle$ (if $b \in \langle a \rangle$ they commute). Then $H = \langle a \rangle$ and $K = \langle b \rangle$ are distinct subgroups of order p, both normal by the first part, and $H \cap K = \{e\}$. Now $aba^{-1}b^{-1} = (aba^{-1})b^{-1} \in K$ and $aba^{-1}b^{-1} = a(ba^{-1}b^{-1}) \in H$, so $aba^{-1}b^{-1} \in H \cap K = \{e\}$, giving $ab = ba$. Hence G is abelian. □

Exercise 13

If a group G has exactly one subgroup H of order k, prove that H is normal in G.

Solution

Let $g \in G$. The set gHg^{-1} is a subgroup of G (it contains e, and $(ghg^{-1})(gh'g^{-1}) = g(hh')g^{-1}$, $(ghg^{-1})^{-1} = gh^{-1}g^{-1}$), and $h \mapsto ghg^{-1}$ is a bijection $H \to gHg^{-1}$, so $|gHg^{-1}| = k$. Since H is the only subgroup of order k, $gHg^{-1} = H$. Thus H is normal. □

Exercise 14

Prove or disprove: $\mathbb{Q}/\mathbb{Z} \cong \mathbb{Q}$.

Solution

This is false. In $\mathbb{Q}/\mathbb{Z}$ the element $\frac{1}{2} + \mathbb{Z}$ is not the identity (since $\frac{1}{2} \notin \mathbb{Z}$) but $2\left(\frac{1}{2} + \mathbb{Z}\right) = 1 + \mathbb{Z} = \mathbb{Z}$, so it has order 2. In $\mathbb{Q}$, however, if $q \neq 0$ then $nq \neq 0$ for every $n \geq 1$, so no non-identity element of $\mathbb{Q}$ has finite order. An isomorphism preserves orders of elements, so $\mathbb{Q}/\mathbb{Z} \ncong \mathbb{Q}$.

Exercise 15

Let G be a finite group and N a normal subgroup of G. If H is a subgroup of G/N, prove that $\phi^{-1}(H)$ is a subgroup in G of order $|H| \cdot |N|$, where $\phi : G \to G/N$ is the canonical homomorphism.

Solution

Subgroup. $\phi(e) = N$ is the identity of G/N, so $e \in \phi^{-1}(H)$. If $a, b \in \phi^{-1}(H)$, then $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} \in H$, since H is a subgroup. So $ab^{-1} \in \phi^{-1}(H)$.

Order. We have $\phi^{-1}(H) = \{g \in G : gN \in H\}$, which is exactly the union of those cosets gN that belong to H. There are $|H|$ such cosets, they are pairwise disjoint, and each has $|N|$ elements. Hence

$$|\phi^{-1}(H)| = |H| \cdot |N|. \quad \square$$

Exercise 16

Let G_1 and G_2 be groups, and let H_1 and H_2 be normal subgroups of G_1 and G_2 respectively. Let $\phi : G_1 \to G_2$ be a homomorphism. Show that ϕ induces a natural homomorphism $\bar{\phi} : G_1/H_1 \to G_2/H_2$ if $\phi(H_1) \subseteq H_2$.

Solution

Define $\bar{\phi}(gH_1) = \phi(g)H_2$.

Well defined. If $gH_1 = g'H_1$, then $g^{-1}g' \in H_1$, so

$$\phi(g)^{-1}\phi(g') = \phi(g^{-1}g') \in \phi(H_1) \subseteq H_2,$$

which means $\phi(g)H_2 = \phi(g')H_2$.

Homomorphism.

$$\bar{\phi}\big((gH_1)(g'H_1)\big) = \bar{\phi}(gg'H_1) = \phi(gg')H_2 = \phi(g)\phi(g')H_2 = \big(\phi(g)H_2\big)\big(\phi(g')H_2\big). \quad \square$$

Exercise 17

If H and K are normal subgroups of G and $H \cap K = \{e\}$, prove that G is isomorphic to a subgroup of $G/H \times G/K$.

Solution

Define $\psi : G \to G/H \times G/K$ by $\psi(g) = (gH, gK)$. It is a homomorphism, since

$$\psi(gg') = (gg'H, gg'K) = (gH \cdot g'H,\ gK \cdot g'K) = \psi(g)\psi(g').$$

If $g \in \ker\psi$, then $gH = H$ and $gK = K$, so $g \in H \cap K = \{e\}$. Thus $\ker\psi$ is trivial, and by the First Isomorphism Theorem $G \cong \psi(G)$, a subgroup of $G/H \times G/K$. $\square$

Exercise 18

Let $\phi : G_1 \to G_2$ be a surjective group homomorphism. Let H_1 be a normal subgroup of G_1 and suppose that $\phi(H_1) = H_2$. Prove or disprove that $G_1/H_1 \cong G_2/H_2$.

Solution

This is false. Let $\phi : \mathbb{Z} \to \mathbb{Z}_2$ be reduction mod 2 (a surjective homomorphism) and $H_1 = \{0\}$, so $H_2 = \phi(H_1) = \{0\}$. Then $G_1/H_1 \cong \mathbb{Z}$ is infinite, while $G_2/H_2 \cong \mathbb{Z}_2$ is finite, so they are not isomorphic.

Remark. The natural map $\bar{\phi} : G_1/H_1 \to G_2/H_2$ from Exercise 16 is always surjective here, and its kernel is $H_1 \ker\phi/H_1$. So it is an isomorphism exactly when $\ker\phi \subseteq H_1$; in the example above, $\ker\phi = 2\mathbb{Z} \not\subseteq \{0\}$.

Exercise 19

Let $\phi : G \to H$ be a group homomorphism. Show that ϕ is one-to-one if and only if $\phi^{-1}(e) = \{e\}$.

Solution

($\Rightarrow$) Suppose ϕ is one-to-one. Since $\phi(e) = e$, we have $e \in \phi^{-1}(e)$. If $\phi(g) = e = \phi(e)$, then $g = e$ by injectivity. So $\phi^{-1}(e) = \{e\}$.

($\Leftarrow$) Suppose $\phi^{-1}(e) = \{e\}$ and $\phi(a) = \phi(b)$. Then

$$\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = e,$$

so $ab^{-1} \in \phi^{-1}(e) = \{e\}$, which gives $a = b$. Hence ϕ is one-to-one. □

Exercise 20

Given a homomorphism $\phi : G \to H$ define a relation $\sim$ on G by $a \sim b$ if $\phi(a) = \phi(b)$ for $a, b \in G$. Show this relation is an equivalence relation and describe the equivalence classes.

Solution

Equivalence relation. It is reflexive since $\phi(a) = \phi(a)$. It is symmetric since $\phi(a) = \phi(b)$ implies $\phi(b) = \phi(a)$. It is transitive since $\phi(a) = \phi(b)$ and $\phi(b) = \phi(c)$ imply $\phi(a) = \phi(c)$.

Equivalence classes. For $a, b \in G$,

$$b \sim a \iff \phi(a) = \phi(b) \iff \phi(a^{-1}b) = e \iff a^{-1}b \in \ker\phi \iff b \in a\ker\phi.$$

So the equivalence class of a is

$$[a] = \phi^{-1}(\phi(a)) = a\ker\phi,$$

the left coset of the kernel containing a. (Since $\ker\phi$ is normal, this is also the right coset $(\ker\phi)a$.) The classes are the fibers of ϕ, and they are in bijection with the elements of the image $\phi(G)$, which matches the isomorphism $G/\ker\phi \cong \phi(G)$.

Additional Exercises: Automorphisms

Throughout, $i_g(x) = gxg^{-1}$, and $U(n)$ is the group of units of $\mathbb{Z}_n$ under multiplication mod n.

Exercise 1

Let $\mathrm{Aut}(G)$ be the set of all automorphisms of G; that is, isomorphisms from G to itself. Prove this set forms a group and is a subgroup of the group of permutations of G; that is, $\mathrm{Aut}(G) \leq S_G$.

Solution. Every automorphism is a bijection $G \to G$, so $\mathrm{Aut}(G) \subseteq S_G$. We check the subgroup criteria.

- The identity map is an automorphism, so $\mathrm{Aut}(G) \neq \emptyset$.
- If $\phi, \psi \in \mathrm{Aut}(G)$, then $\phi \circ \psi$ is a bijection, and $(\phi \circ \psi)(xy) = \phi(\psi(x)\psi(y)) = \phi(\psi(x))\phi(\psi(y))$. So $\phi \circ \psi \in \mathrm{Aut}(G)$.
- If $\phi \in \mathrm{Aut}(G)$, then ϕ^{-1} is a bijection. For $x, y \in G$, write $x = \phi(a)$ and $y = \phi(b)$. Then $\phi^{-1}(xy) = \phi^{-1}(\phi(ab)) = ab = \phi^{-1}(x)\phi^{-1}(y)$. So $\phi^{-1} \in \mathrm{Aut}(G)$.

Hence $\mathrm{Aut}(G) \leq S_G$, and in particular $\mathrm{Aut}(G)$ is a group under composition. □

Exercise 2

An inner automorphism of G, $i_g : G \to G$, is defined by $i_g(x) = gxg^{-1}$ for $g \in G$. Show that $i_g \in \mathrm{Aut}(G)$.

Solution. i_g is a homomorphism:

$$i_g(xy) = gxyg^{-1} = (gxg^{-1})(gyg^{-1}) = i_g(x)i_g(y).$$

It is bijective, with inverse $i_{g^{-1}}$: $i_{g^{-1}}(i_g(x)) = g^{-1}gxg^{-1}g = x$ and $i_g(i_{g^{-1}}(x)) = gg^{-1}xgg^{-1} = x$. So $i_g \in \mathrm{Aut}(G)$. □

Exercise 3

The set of all inner automorphisms is denoted by $\mathrm{Inn}(G)$. Show that $\mathrm{Inn}(G)$ is a subgroup of $\mathrm{Aut}(G)$.

Solution. For $g, h \in G$ and $x \in G$,

$$(i_g \circ i_h)(x) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = i_{gh}(x), \qquad \text{so} \qquad i_g \circ i_h = i_{gh}. \tag{$*$}$$

Then:

- $i_e = \mathrm{id}$, so $\mathrm{id} \in \mathrm{Inn}(G)$.
- By $(*)$, $i_g \circ i_h = i_{gh} \in \mathrm{Inn}(G)$.
- $i_g^{-1} = i_{g^{-1}} \in \mathrm{Inn}(G)$ (Exercise 2).

Hence $\mathrm{Inn}(G) \leq \mathrm{Aut}(G)$. □

Exercise 4

Find an automorphism of a group G that is not an inner automorphism.

Solution. Let $G = \mathbb{Z}_3$ and $\phi(x) = -x$ (that is, $\phi(0) = 0$, $\phi(1) = 2$, $\phi(2) = 1$). Then ϕ is a homomorphism, since $-(x + y) = -x + -y$ in an abelian group, and it is bijective (it is its own inverse), so $\phi \in \mathrm{Aut}(\mathbb{Z}_3)$. In an abelian group, $i_g(x) = g + x - g = x$ for every g, so $\mathrm{Inn}(\mathbb{Z}_3) = \{\mathrm{id}\}$. But $\phi(1) = 2 \neq 1$, so ϕ is not inner.

(The same works for $\mathbb{Z}$ with $n \mapsto -n$.)

Exercise 5

Let G be a group and i_g an inner automorphism of G, and define a map $G \to \mathrm{Aut}(G)$ by $g \mapsto i_g$. Prove that this map is a homomorphism with image $\mathrm{Inn}(G)$ and kernel $Z(G)$. Use this result to conclude that $G/Z(G) \cong \mathrm{Inn}(G)$.

Solution. Let $\Phi(g) = i_g$. By $(*)$ in Exercise 3, $\Phi(gh) = i_{gh} = i_g \circ i_h = \Phi(g)\Phi(h)$, so Φ is a homomorphism.

Image. $\Phi(G) = \{i_g : g \in G\} = \mathrm{Inn}(G)$ by definition.

Kernel. $g \in \ker \Phi$ if and only if $i_g = \mathrm{id}$, if and only if $gxg^{-1} = x$ for all $x \in G$, if and only if $gx = xg$ for all $x \in G$, if and only if $g \in Z(G)$.

By the First Isomorphism Theorem, $G/\ker \Phi \cong \Phi(G)$, that is,

$$G/Z(G) \cong \mathrm{Inn}(G). \quad \square$$

Exercise 6

Compute $\mathrm{Aut}(S_3)$ and $\mathrm{Inn}(S_3)$. Do the same thing for D_4.

Solution.

S_3. We have $Z(S_3) = \{(1)\}$ (Chapter 10, Exercise 14(a)), so by Exercise 5, $\mathrm{Inn}(S_3) \cong S_3/\{(1)\} \cong S_3$, a group of order 6.

Now $S_3 = \langle (123), (12) \rangle$, so an automorphism ϕ is determined by $\phi((123))$ and $\phi((12))$. Automorphisms preserve orders, so $\phi((123)) \in \{(123), (132)\}$ (2 choices) and $\phi((12)) \in \{(12), (13), (23)\}$ (3 choices). Hence $|\mathrm{Aut}(S_3)| \leq 6$. Since $\mathrm{Inn}(S_3) \leq \mathrm{Aut}(S_3)$ already has order 6,

$$\mathrm{Aut}(S_3) = \mathrm{Inn}(S_3) \cong S_3.$$

D_4. Write $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$.

Inner automorphisms. $Z(D_4) = \{1, r^2\}$, so $\mathrm{Inn}(D_4) \cong D_4/\langle r^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (Chapter 10, Exercise 2), of order 4. Using $rs = sr^{-1}$, we get $i_r(r) = r$, $i_r(s) = rsr^{-1} = sr^2$, and $i_s(r) = r^{-1}$, $i_s(s) = s$. So

$$\mathrm{Inn}(D_4) = \{\mathrm{id},\ i_r,\ i_s,\ i_{sr} = i_s \circ i_r\},$$

where, writing each map as (image of r, image of s), these are (r, s), (r, sr^2), (r^{-1}, s), (r^{-1}, sr^2).

All automorphisms. An automorphism ϕ is determined by $\phi(r)$ and $\phi(s)$. Since $\phi(r)$ has order 4, $\phi(r) = r^{\pm 1}$ (2 choices). Since $\phi(s)$ has order 2 and must not lie in $\langle r \rangle$ (otherwise the image would not be all of D_4), $\phi(s) \in \{s, sr, sr^2, sr^3\}$ (4 choices). So $|\mathrm{Aut}(D_4)| \leq 8$. Conversely, each of these 8 choices satisfies the defining relations: if $\phi(r) = r^{\varepsilon}$ and $\phi(s) = sr^j$, then

$$\phi(s)\phi(r)\phi(s) = sr^j\, r^{\varepsilon}\, sr^j = r^{-(j+\varepsilon)} r^j = r^{-\varepsilon} = \phi(r)^{-1},$$

so it defines a homomorphism, which is onto (its image contains r and s) and hence bijective. Thus $|\mathrm{Aut}(D_4)| = 8$.

To identify the group, let $\alpha : r \mapsto r,\ s \mapsto sr$ (so $\alpha^k(s) = sr^k$, and α has order 4) and $\beta : r \mapsto r^{-1},\ s \mapsto s$ (order 2). Then $\beta\alpha\beta(r) = r$ and $\beta\alpha\beta(s) = \beta(\alpha(s)) = \beta(sr) = sr^{-1} = \alpha^{-1}(s)$, so $\beta\alpha\beta = \alpha^{-1}$. Since $\beta \notin \langle \alpha \rangle$ (as $\beta(r) = r^{-1} \neq r = \alpha^k(r)$), the subgroup $\langle \alpha, \beta \rangle$ has order 8, so it is all of $\mathrm{Aut}(D_4)$, and

$$\mathrm{Aut}(D_4) \cong D_4.$$

Here $\mathrm{Inn}(D_4)$ is a subgroup of index 2 in $\mathrm{Aut}(D_4)$, so $\mathrm{Aut}(D_4) \neq \mathrm{Inn}(D_4)$.

Exercise 7

Find all of the homomorphisms $\phi : \mathbb{Z} \to \mathbb{Z}$. What is $\mathrm{Aut}(\mathbb{Z})$?

Solution. A homomorphism ϕ is determined by $k = \phi(1)$, since $\phi(n) = n\phi(1) = kn$. Conversely, for each $k \in \mathbb{Z}$ the map $n \mapsto kn$ is a homomorphism. So the homomorphisms $\mathbb{Z} \to \mathbb{Z}$ are exactly $\phi_k(n) = kn$, $k \in \mathbb{Z}$.

ϕ_k is bijective if and only if $k\mathbb{Z} = \mathbb{Z}$ and $k \neq 0$, that is, $k = \pm 1$. Hence

$$\mathrm{Aut}(\mathbb{Z}) = \{\mathrm{id},\ n \mapsto -n\} \cong \mathbb{Z}_2.$$

Exercise 8

Find all of the automorphisms of $\mathbb{Z}_8$. Prove that $\mathrm{Aut}(\mathbb{Z}_8) \cong U(8)$.

Solution. A homomorphism $\phi : \mathbb{Z}_8 \to \mathbb{Z}_8$ is determined by $k = \phi(1)$, since $\phi(a) = ka$. Every $k \in \mathbb{Z}_8$ gives a homomorphism $\phi_k(a) = ka$ (Exercise 9 below). By Exercise 10 below, ϕ_k is an

automorphism if and only if k generates $\mathbb{Z}_8$, that is, $\gcd(k, 8) = 1$. Hence the automorphisms of $\mathbb{Z}_8$ are

$$\phi_1 = \mathrm{id}, \quad \phi_3, \quad \phi_5, \quad \phi_7,$$

so $|\operatorname{Aut}(\mathbb{Z}_8)| = 4$. Since $\phi_k \circ \phi_l(a) = k(la) = (kl)a = \phi_{kl}(a)$, the map $U(8) \to \operatorname{Aut}(\mathbb{Z}_8)$, $k \mapsto \phi_k$, is a homomorphism. It is injective ($\phi_k(1) = k$) and surjective (by the list above), so

$$\operatorname{Aut}(\mathbb{Z}_8) \cong U(8) = \{1, 3, 5, 7\}.$$

(Both are isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, since $3^2 = 5^2 = 7^2 = 1$ in $U(8)$.) This is the case $n = 8$ of Exercise 12.

Exercise 9

For $k \in \mathbb{Z}_n$, define a map $\phi_k : \mathbb{Z}_n \to \mathbb{Z}_n$ by $a \mapsto ka$. Prove that ϕ_k is a homomorphism.

Solution. Multiplication in $\mathbb{Z}_n$ is well defined and distributes over addition, so for $a, b \in \mathbb{Z}_n$,

$$\phi_k(a + b) = k(a + b) = ka + kb = \phi_k(a) + \phi_k(b). \quad \square$$

Exercise 10

Prove that ϕ_k is an isomorphism if and only if k is a generator of $\mathbb{Z}_n$.

Solution. The image of ϕ_k is $\{ka : a \in \mathbb{Z}_n\} = \langle k \rangle$.

($\Rightarrow$) If ϕ_k is an isomorphism, it is onto, so $\langle k \rangle = \mathbb{Z}_n$, that is, k is a generator.

($\Leftarrow$) If k generates $\mathbb{Z}_n$, then $\langle k \rangle = \mathbb{Z}_n$, so ϕ_k is onto. A surjective map from a finite set to itself is also injective, so ϕ_k is a bijective homomorphism, that is, an isomorphism. $\square$

Exercise 11

Show that every automorphism of $\mathbb{Z}_n$ is of the form ϕ_k, where k is a generator of $\mathbb{Z}_n$.

Solution. Let $\phi \in \operatorname{Aut}(\mathbb{Z}_n)$ and put $k = \phi(1)$. For $a \in \mathbb{Z}_n$,

$$\phi(a) = \phi(a \cdot 1) = a\,\phi(1) = ka = \phi_k(a),$$

so $\phi = \phi_k$. Since ϕ_k is an isomorphism, k is a generator of $\mathbb{Z}_n$ by Exercise 10. $\square$

Exercise 12

Prove that $\psi : U(n) \to \operatorname{Aut}(\mathbb{Z}_n)$ is an isomorphism, where $\psi : k \mapsto \phi_k$.

Solution. Recall that $U(n) = \{k \in \mathbb{Z}_n : \gcd(k, n) = 1\}$ is exactly the set of generators of $\mathbb{Z}_n$.

- *Well defined:* for $k \in U(n)$, $\phi_k \in \operatorname{Aut}(\mathbb{Z}_n)$ by Exercise 10.
- *Homomorphism:* $(\phi_k \circ \phi_l)(a) = k(la) = (kl)a = \phi_{kl}(a)$, so $\psi(kl) = \psi(k)\psi(l)$.
- *Injective:* if $\phi_k = \phi_l$, then $k = \phi_k(1) = \phi_l(1) = l$.
- *Surjective:* by Exercise 11, every automorphism of $\mathbb{Z}_n$ is ϕ_k for some generator $k \in U(n)$.

Hence ψ is an isomorphism, and $U(n) \cong \operatorname{Aut}(\mathbb{Z}_n)$. $\square$